

RESEARCH ARTICLE

An AI Driven, Reliability-Based Planning Framework for Electric Vehicles Integrated Distribution Network Considering Renewable Energy Support

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ABSTRACT The presence of Electric Vehicle Charging Infrastructure (EVCI) in the Distribution Network (DN) adversely impacts grid operations, affecting system voltage profile, power loss, and reliability, particularly under unplanned placement. To address this, a comprehensive planning framework is proposed considering EVCI, Photovoltaic (PV), and Battery Storage System (BSS), while minimizing techno-economic and reliability impacts using a novel Enhanced Harris Hawk Optimization (EHHO) technique that adheres to system constraints. The research employs a Deep Learning-based Gated Recurrent Unit algorithm to accurately forecast EV charging load and PV generation. The EVCI is allocated in the DN using a novel Bus Reliability Index (BRI) and load density method. For more practicality, a time dependent failure rate is considered for feeder, PV and BSS to model the reliability. The proposed methodology is tested on the IEEE 15-bus DN and the 28-bus Indian rural DN. Comparative performance analysis demonstrates the superiority of the proposed approach. Results show that integrating PV and BSS with EVCI significantly improves the system reliability, voltage profile and reduces power loss. Specifically, in terms of reliability index Expected Energy Not Served (EENS), the presence of PV and BSS reduces EENS by 16% in the 15-bus DN and by 27% in the 28-bus DN compared to integration with only EVCI.

INDEX TERMS System average interruption frequency index, system average interruption duration index, average expected energy not served, expected energy not served, enhanced HHO, distribution network.

I. INTRODUCTION

Distribution Networks (DNs) face substantial challenges that potentially threaten both their operational efficiency and long-term sustainability. These challenges are mostly connected to rising electricity demand due to increasing EV adoption, increasing integration of RES, and aging infrastructures, among others. Managing these complexities requires innovative strategies to maintain reliable operations [1]. In the

wake of buttressing global efforts to combat climate change, the integration of Electric Vehicle Charging Infrastructures (EVCI), Photovoltaic (PV), and Battery Storage System (BSS) has become increasingly important in modern DN [2], [3]. The rise in EV adoption requires more EVCI incorporation in the DN, presenting both opportunities and challenges for Distribution System Operators (DSOs).

One of the most concerning challenges faced with the integration of EVCI and PV in the DN is their inherent operational uncertainty [4]. Therefore, adopting a robust approach to tackle this uncertainty during the DN planning

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phase is vital. EV charging behavior is profoundly stochastic, which is influenced by charging duration, arrival times, and battery's State of Charge (SOC) [5]. These complexities present challenges in accurately forecasting charging loads, adversely impacting the DN's loading. As a means of mitigating the associated EV charging load uncertainties, many authors have adopted several forecasting methods, including conventional methods such as two-point estimation (2m point) [6], Auto-regression Integrated Moving Average (ARIMA) [7], as well as advanced methods such as Seasonal Auto-regression Integrated Moving Average (SARIMA) [8], Monte Carlo Simulations (MCS) [9], [10], Machine Learning methods [11], and Deep Learning (DL) methods [12]. Table 1 provides a comprehensive description of the advantages and disadvantages of these algorithms. The comparison shows that classical models like ARIMA and SARIMA are simple, interpretable, and suitable for small or stable datasets but fail to capture non-linear and highly uncertain EV charging patterns. MCS effectively models uncertainty and variability but lacks precision as it is not data-driven. In contrast, advanced models such as LSTM, GRU, and ensemble learning handle large datasets, non-linearity, and long-term dependencies more accurately, making them better suited for realistic EV load forecasting, though they require high computation and careful tuning. However, the GRU technique features a simplified architecture with fewer parameters, which enables faster training compared to LSTM.

ABBREVIATIONS

Symbol	Description		
Δh	Charging time interval.	$E_{BSS}^{min}/E_{BSS}^{max}$	Minimum/Maximum limit of the BSS energy.
ΔL_{EVCI}	Charging load after EVCI integration.	$EENS_i$	Expected Energy Not Served at the individual bus i .
λ_k	Failure rate of k^{th} component (i.e., PV and BSS) in the network.	$EENS_{Max}$	System's maximum EENS value.
$\lambda_i, \lambda_{avg,j}$	Failure rate of bus i , average failure rate of the line j .	$E_{BSS}^h, E(h-1)_{BSS}$	BSS energy at time $h, h-1$.
λ_{EVCI}	Line failure rate after EVCI integration.	E_{in}	Initial energy of the target in HHO.
α_1^F, α_2^F	Shape parameter of Weibull Distribution for feeder.	E_{conv}, E^M, E^I	Escaping energy in Traditional HHO, MHHO, IHHO.
β_2^F	Scale parameter of Weibull Distribution for feeder.	fit, fit_{best}	Current fitness and best fitness values of IHHO.
η_{chg}/η_{dischg}	Charging/ discharging efficiency of BSS.	h_{t-1}	Hidden state from the previous time step (GRU's memory).
σ/ϕ_h	Sigmoid/tanh activation functions.	$\hat{h}_t/h_t$	Candidate's hidden state/Final hidden state (or output) at time step t .
$b_z/b_r/b_h$	Update gate bias /Reset gate bias/ candidate hidden state vectors of GRU.	HOC	Hourly Operational Cost.
C_{PV}	PV cost.	it, it_{max}	Current iteration, maximum number of iterations in optimization technique.
C_{P-loss}	Cost of power loss.	$InvstC_{BSS}, InvstC_{PV}$	Investment cost of BSS, PV.
C_{Add}	Engineering cost.	IF/IR	Inflation rate/ Interest rate.
$COST_{Total}$	Total cost.	J	Jump strength of the target in HHO.
CCF	Cost Conversion Factor.	k	Number of elements in the network.
C_M	Maintenance cost.	K_s	EV charging simultaneity constant.
C_{Elec}	Cost of Electricity.	L_i	Active load at the i^{th} number of buses.
D	Number of days in a year.	L_{base}	Base load.
E_{BSS}^h	Battery energy stores at time h .	L_{total}/L_{zone}	Total load of the entire network /Total load in each zone of the network.
		m_k, η_k	Weibull shape and scale parameters of component k (i.e., PV and BSS)..
		nPop	Number of populations.
		N	Total number of data samples in GRU.
		N_i, N_k	Number of customers on the i^{th} number of buses, total number of k^{th} components.
		N_{PV}/BSS	Number of PV panels/BSS units.
		N_{EV}	Number of EVs in the planning area.
		N_y	Number of planning years.
		P_{ch}	The charger power rating.
		$P_G(h)$	Total generating power at time h .
		$P_D(h)$	Power drawn by the residential load on the grid at time t .
		P_{PV}/P_{BSS}	Nominal power of PV/BSS.
		P_{loss}^h	Power loss at hour h .
		P_{chg}^h/P_{dischg}^h	Charging/Discharging power of the BSS at time t .
		$P_{BSS}^{max}/P_{BSS}^{min}$	Maximum/Minimum power of the BSS.
		$r_{j,k}, r_{avg,j}$	Outage time at j^{th} line of k^{th} element, Average outage time of line j .

SOC_{min}/SOC_{max}	Minimum/maximum state of Charge of the battery.
S_{CI}	CI capacity.
C_{BSS}	Total Cost of BSS.
U_i	Unavailability of the i^{th} number of buses.
U_{base}/U_{EVCI}	Unviability of the bus before/after EVCI integration.
U_z	The weight matrix applied to the previous hidden state h_{t-1} .
ub, lb	Upper bound and lower bound of the particle in optimization.
V_i^h	The voltage at bus i at time h .
V_{max}/V_{min}	Maximum/Minimum voltage.
W_z	The weight matrix applied to the current input x_t .
X_{std}	The standardized value of the dataset in GRU.
$x/\hat{y}$	Actual/Predicted values of the input dataset in GRU.
$x_{max} - x_{min}$	The actual values range in GRU.
$X(it), X(it+1)$	Current position of hawk at iteration it , updated position of hawk in the next iteration ($it+1$).
$X_{rd}(it), X_{tar}(it), X_m(it)$	Random position of hawk, position of target, average hawk position at iteration it .
z_t/r_t	Update gate output vector at time step t / The reset gate output vector at time t in GRU.

PV power generation is inherently intermittent [2], as it is hanged on solar irradiance and weather conditions. This uncertainty in PV output impacts the DN operational stability and technical performance. The integration of BSS with PV in the DN mitigates PV intermittency, ensuring continuous and reliable power supply [3]. During PV downtime, BSS supports the grid by meeting demand and maintaining stability.

In the wake of these challenges, a comprehensive investigation of the network reliability is important, especially when EVCI, PV, and BSS are integrated into the grid. Their presence in the DN significantly impacts the network performance, resulting in customer dissatisfaction due to power supply disruptions [13]. Additionally, the inherent uncertainty in EV charging load and PV power generation presents a major threat to network reliability and stability. A robust reliability evaluation is essential, as network reliability affects electricity quality for residential, commercial, and industrial customers, reduces outages, and ensures continuous supply [14]. Reliability study also provides the DSO valuable insight in determining the suitable allocation of EVCI in the grid, and evaluating the operational state of the system.

Table 2 presents a comparative review of the existing literature on EVCI planning. Some studies have proposed the optimal allocation of Charging Infrastructures (CIs) within the DN using metaheuristic algorithms, typically considering cost as the objective function [15], [16]. Similarly, the authors in [17] focus on the optimal allocation of EV and PV in the DN, prioritizing voltage profile improvement as the fitness function. In [18], the authors aim to minimize the network power loss while simultaneously allocating EV and PV in the DN. Despite the significant contributions made in [15], [16], [17], and [18], these studies did not incorporate BSS to mitigate the challenges associated with renewable energy intermittency. In [19], the authors proposed an AI based optimal planning of EVCS and DG considering reliability analysis. However, the authors assumed a constant component failure rate, which is unrealistic since the probability of aging-related failures increases over time.

Further, the assessments in these studies ignored the impact of the grid reliability in their optimization model, which could potentially cause instability and negatively influence system performance. Unlike other studies [15], [16], [17], [18], the present work differs by analyzing the grid reliability considering the presence of EV, PV, and BSS in the network. A novel methodology is adopted to properly allocate EVCI in the DN, reducing their adverse technical impacts on the DN.

Many studies have employed optimization using metaheuristics and other algorithms to place EV and PV in the DN [19], [20]. The problem with these techniques is that they often concentrate CI deployment in consecutive buses in the DN, which initiates load imbalance and creates inconveniences for both EV owners and DNOs.

Moreover, soft computing approaches are known for their adaptability, robustness, and efficiency in solving complex real-world optimization problems. As shown in Table 2, in [21], [22], and [23], conventional methods such as Particle Swarm Optimization (PSO), Grey Wolf Optimization, Genetic Algorithm (GA) were applied to identify the optimal location and size of PV panel. However, these methods often encounter challenges, including difficulties in parameter tuning, slower convergence, and limited robustness. To overcome these shortcomings, the present study develops an Enhanced Harris Hawks Optimization (EHHO) technique.

This study developed a novel index based on reliability that is used to judiciously allocate the CI in the DN. The network is divided into zones to effectively address the uneven deployment of EV charging loads and avoid the clustering of CI in consecutive buses on the grid. The present work adopts the deep learning Gated Recurrent Unit (GRU) to forecast the EV charging loads and PV generation using real world historic data. Furthermore, PV and BSS are optimally integrated into the network utilizing the novel EHHO algorithm. The reliability and technical performance of the DN are evaluated while identifying critical locations for potential improvement. Key grid reliability indices, including the System Average Interruption Frequency Index (SAIFI),

TABLE 1. Summary of EV Charging Demand Uncertainty Modelling Algorithms.

Algorithms/Ref.	Advantage	Disadvantage
2m point [6]	- With a large data, the algorithm can identify temporal patterns within the data. - Effective in learning non-linearity in the EV dataset. - Outliers such as holidays or abnormal charging sessions are modeled using this algorithm.	- Requires a large data for both training and validation, needing large memory and processing power. - Longer training and hyperparameter tuning due to a large data. - Pre-processing of the data is resource-intensive.
ARIMA [7]	- The algorithm application is straightforward, easing the mathematical interpretation of the predicted result. - ARIMA can work with small data, suitable for data-scarce conditions - Under stable and non-seasonal data patterns, ARIMA performs short-term load forecasting	- ARIMA captures linear relationships, reducing its efficiency when working with non-linear problems. - ARIMA struggles with the instability in EV loading pattern (i.e., user behavior, etc). - ARIMA is designed for univariate (single variable) forecasting, not effective for multiple variables.
SARIMA [8]	- Accurately models EV charging data, capturing repeated patterns including daily, weekly, or seasonal cycles (i.e., weekdays and weekends). - Accurately perform forecasting of EV charging load for a short period. - The algorithm provides a clear interpretation of forecasting results.	- Incapable of modeling nonlinear patterns of EV charging load, such as user behavior, etc. - The algorithm is ineffective with large datasets. - SARIMA parameters are complex than ARIMA, making the tuning difficult and time-consuming during forecasting.
MCS [9], [10]	- MCS is suited for capturing uncertainty in the EV user behavior, arrival times, and charging times. - MCS is flexible in modelling nonlinear data. - MCS delivers probability distributions of possible results instead of single-point forecasts, valuable for grid planning.	- MCS is not data-driven, meaning it doesn't learn from data patterns but is based on predefined rules. - MCS has limited precision in forecasting. - Due to the probabilistic results of MCS, it is difficult to validate the result against real data using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), etc.
Ensemble Learning Model [11]	- This method improves forecasting accuracy by combining multiple methods. - Ensemble models learn nonlinearity and complex patterns, which is common with EV charging data due to diverse charging patterns.	- Training and evaluation processes are resource-intensive, particularly for large data. - Ensemble models are often seen as closed-box model, making them less interpretable than ARIMA.
Long Short-Term Memory (LSTM) [12]	- LSTM model remembers patterns over long sequences, making it ideal for long-term time-series forecasting. - LSTM is suitable for real-time EV load forecasting.	- The LSTM model requires substantial training data. - LSTM performance depends on tuning (number of layers, units, learning rate), which is complex and time-consuming.
GRU	- GRU effectively learns both short and long-term dependencies in sequential EV charging load data. - With fewer gates and parameters, GRU trains faster using less memory and delivers accurate results for forecasting.	Effectiveness can depend on the nature of the dataset and task.

System Average Interruption Duration Index (SAIDI), Expected Energy Not Served (EENS), and Average Expected Energy Not Served (AEENS), are considered in the

assessment [24]. To accurately assess the effect of failure rates on the DN, time-dependent failure rates of the feeder, PV, and BSS are modeled. The results of EHHO

TABLE 2. Summary of literature review on EV, PV, and BSS allocation in the DN.

Ref.	Integrated Alternatives						Objective Function		Reliability Analysis	EV distribution by Zones	Solution method
	EVCi		PV		BSS		Technical	Economic			
	Size	Location	Size	Location	Size	Location					
[18]	✓	✓	✓	✓	✗	✗	✓	✗	✓	✗	Hybrid of grey wolf optimization and particle swarm optimization
[20]	✓	✓	✓	✓	✓	✓	✓	✓	✗	✗	PSO
[21]	✗	✗	✓	✓	✗	✗	✓	✓	✗	✗	Grey Wolf Optimization
[22]	✗	✗	✓	✓	✓	✓	✗	✓	✗	✗	Genetic Algorithms
[25]	✓	✓	✓	✓	✗	✗	✓	✓	✗	✗	Dragonfly algorithm
[26]	✗	✗	✓	✓	✗	✗	✓	✗	✗	✗	Dingo Optimization Algorithm
[27]	✗	✗	✓	✓	✗	✗	✓	✗	✗	✗	Improved Equilibrium Optimizer
Present Work	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	EHHO

and all other techniques adopted in this present work are compared with other prominent algorithms to validate their performance. The results of various comparisons demonstrate excellent performance of the adopted algorithms. The primary contributions of this paper are as follows:

- Considering the intermittency of EV and PV, a comprehensive system reliability-based planning model of DN is developed to minimize power loss cost, installation, operation and maintenance cost of PV and BSS, while satisfying the security constraints.
- A novel EHGO technique is developed with enhancements in escaping energy evaluation and the exploration phase of traditional HGO, which collectively improve solution diversity, global search capability, and convergence speed. To prove the effectiveness of EHGO, standard benchmark functions are tested and results are compared with other techniques.
- Utilizing a novel reliability-oriented Bus reliability Index (BRI) and load density method, EVCIs are allocated in the DN. To enhance practical applicability, the model incorporates time-dependent failure rates of the feeder, PV system, and BSS in evaluating reliability indices.
- A DL-based GRU model is employed to effectively capture the uncertainties of EV load and PV generation through accurate forecasting. To ensure practical relevance, the model is trained and validated using real-world historical EV and PV data.

II. RELIABILITY INDICES

Overloading of the network under unplanned CI deployment conditions deteriorates the DN reliability, which affects the system operation, causing customer dissatisfaction. In this study, reliability assessment highlights two groups of indices. The first one is the customer-based indices, which comprise SAIFI and SAIDI. On the other hand, the second one is the load-based indices, which include EENS and AEENS.

A. SYSTEM AVERAGE INTERRUPTION FREQUENCY INDEX (SAIFI)

SAIFI quantifies the frequency customers encounter outages or interruptions within a specified period [28], [29]. SAIFI performance declines as the number of interruptions increases, along with the total number of customers in the system [30]. SAIFI can be calculated as follows.

$$SAIFI = \frac{\sum \lambda_i N_i}{\sum N_i} \text{ interruptions/customer/year} \quad (1)$$

B. SYSTEM AVERAGE INTERRUPTION DURATION INDEX (SAIDI)

SAIDI determines the period of interruption experienced by each customer in the system. It reflects the interruption length and the number of customers [30]. SAIDI is calculated using the equation below:

$$SAIDI = \frac{\sum U_i N_i}{\sum N_i} \text{ hours/customer/year} \quad (2)$$

C. AVERAGE EXPECTED ENERGY NOT SERVED (AEENS)

AEENS is the total EENS divided by the number of customers that experienced interruption per year. AEENS is built on the load requirements at individual buses in the network. This normalizes the EENS by the total number of customers in the network. Moreover, AEENS denotes the average EENS per the number of customers. This provides valuable insight into the average customer affected by the interruption. To evaluate AEENS, the following equation is used [29], [30].

$$AEENS = \frac{\sum L_i U_i}{\sum N_i} \text{ kWh/customer/year} \quad (3)$$

D. EXPECTED ENERGY NOT SERVED (EENS)

In the reliability assessment of the power system, the EENS metric is widely adopted to quantify the unserved energy during the interruption period [18], [31].

$$EENS = \sum_{i=1}^m U_i L_i \text{ kWh/year} \quad (4)$$

$$U_i = \sum_{j=1}^l \lambda_{avg,j} r_{avg,j} \quad (5)$$

$$\lambda_{avg,j} = \sum_{k=1}^K N_k \times \lambda_{avg,k} \quad (6)$$

$$r_{avg,j} = \frac{\sum_{k=1}^K \lambda_{avg,k} \times r_{j,k}}{\lambda_{avg,j}} \quad (7)$$

1) TIME DEPENDENT FAILURE RATE OF FEEDER

$$\lambda_1^F(h) = \alpha_1^F h_1 \leq h \leq h_2 \quad (8)$$

$$\lambda_2^F(h) = \alpha_1^F + \alpha_2^F \beta_2^F t^{\beta_2^F - 1} \quad (9)$$

$\lambda_1^F(h)$, $\lambda_2^F(h)$ are the failure rate of feeder during stabilization period and aging period, respectively [18].

2) TIME DEPENDENT FAILURE RATE AND UNAVAILABILITY OF SYSTEM AFTER EVCI INTEGRATION

The incorporation of CIs into the DN increases the interruption time of the DN reliability from the point of common coupling, where the CI is connected to the DN. When the CIs are integrated, the failure rate ($\lambda^F(h)$) and unavailability (U_i) are updated to (λ_{EVCI}) and (U_{EVCI}), respectively, using the equations (10) and (11) [18].

$$\lambda_{EVCI}(h) = (L_{bsae} + \Delta L_{EVCI}) \frac{\lambda^F(h)}{L_{base}} \quad (10)$$

$$U_{EVCI}(h) = (L_{base} + \Delta L_{EVCI}) \frac{U_{base}}{L_{base}} \quad (11)$$

3) TIME DEPENDENT FAILURE RATE OF PV AND BSS

The DSO feasibly utilizes PV by locally installing it at the load center. As the PV is integrated into the network, the failure of the system becomes modified as mathematically expressed in (12) [31].

The PV and BSS failure behaviors are assumed to follow a Weibull distribution, which effectively captures aging and wear-out effects. The failure rate for component k ($k \in \{PV,$

BSS}) is given in (12).

$$\lambda_k(h) = \frac{m_k}{\eta_k} \left(\frac{h}{\eta_k} \right)^{m_k - 1} \quad (12)$$

4) BUS RELIABILITY INDEX (BRI)

To determine the reliability of each bus of the DN, a novel index BRI is developed. The BRI of each bus is calculated using as follows.

$$BRI = \frac{EENS_i}{EENS_{Max}} \quad (13)$$

III. MATHEMATICAL MODELLING OF DEVICE, OBJECTIVE FUNCTION, AND TECHNICAL CONSTRAINTS

A detailed formulation of the mathematical model, including the objective function and constraints, is presented below.

A. MATHEMATICAL MODELLING

The mathematical modelling relative to devices that are considered in the proposed planning framework is properly discussed in the sections below.

1) EVCI MODELLING

EVCIs are classified according to charging levels and defined per standards (i.e. SAE, IEC etc). The various EV charging types include AC Level-1, AC Level-2, and DC fast charging (DCFC) [32]. In addition, with the new regulations, DCFC specification includes DC Level-1 and DC Level-2 [33]. Hence, studies have widely adopted AC Level-1 and AC Level-2 as popular options for residential charging purposes.

The uncertainty associated with EV charging demand reduces the accuracy in predicting the expected charging demand and the overall demand pattern of DN. This can negatively influence the grid's technical performance, reliability, and stability. Modelling EV charging demand uncertainty is vital in managing its dynamic charging pattern to ensure efficient operation of the grid. In this study, hourly EV charging demand is forecasted using GRU, a machine-learning-based algorithm.

2) PV MODELING

The uncertainty of solar PV generation reduces the accuracy of predicting renewable power output, affecting EVCS reliability and grid stability. Modeling PV variability is therefore essential for efficient integration with storage and the grid. In the present work, the hourly PV generation is estimated by GRU based forecasting technique.

3) BATTERY STORAGE SYSTEM (BSS)

Addressing the associated challenges posed by the intermittence of the PV output, BSS integration has become a paramount component of modern energy systems. Additionally, this technology supports DSO in addressing issues such as reverse power flow and voltage rise in the Radial DN, which frequently arise when PV panels are incorporated in the DN [31]. In this work, a Lithium-ion battery is considered

for the BSS. The BSS effectively stores extra power generated by the PV and efficiently discharges it when needed to maintain system stability. The equations utilized to measure the charging and discharging energies of the BSS are provided below [31].

Charging Period:

$$E_{BSS}^h = E(h-1)_{BSS} + \eta_{chg} \times P_{chg} \times \Delta h \quad (14)$$

Discharging Period:

$$E_{BSS}^h = E(h-1)_{BSS} - P_{dischg} \times \Delta h / \eta_{dischg} \quad (15)$$

The minimum and maximum limits of the BSS energy are controlled with the following equation:

$$E_{BSS}^{min} \leq E_{BSS}^h \leq E_{BSS}^{max} \quad (16)$$

B. MATHEMATICAL MODELLING OF OBJECTIVE FUNCTION AND TECHNICAL CONSTRAINTS

This section describes the objective functions and the technical constraints employed in the optimal allocation of the PV and BSS in the DN.

1) OBJECTIVE FUNCTION

In this current study, the objective function is developed for the technical and economic benefits of DSO. PV installation, maintenance costs [31], BSS investment cost [34], and Power Loss Cost [35] are minimized to address the technical and economic challenges of the DN. The following are the details of the objective function.

$$\min Cost_{Total} = C_{PV} + Invst C_{BSS} + C_{P-loss} \quad (17)$$

- Cost of PV (C_{PV})

The PV cost covering installation (C_{PV}) operation and maintenance (C_M) is given in (18). CCF given in (19) is the collective current value connected to the costs [31], which represents the cost-related measure used to estimate the total expenses over a specific period. The maintenance cost of PV is 1% of the investment cost [31].

$$C_{PV} = \sum_{n=1}^{N_{PV}} Invst C_{PV} \times P_{PV} + \sum_{n=1}^{N_{PV}} \sum_{h=1}^{24} HOC \times P_{PV} \times N_y \times CCF \times D + C_M \times CCF \times N_y \quad (18)$$

$$CCF = \frac{1 - \left(\frac{1+IF}{1+IR}\right)^{N_y}}{1 - \left(\frac{1+IF}{1+IR}\right)} \quad (19)$$

The value of N_y and D are taken as 10 years and 365 days, respectively in the present work.

- Cost of BSS (C_{BSS})

BSS presence in the DN enables a greater proportion of self-generated energy coming from the PV to be directly utilized during PV idle period, enhancing network performance by minimizing reliance on external energy sources and improving network efficiency, benefiting DSO. This section

carefully evaluates the economic application of BSS deployment in the network. The total cost of the BSS comprises BSS investment cost (BSS cost), maintenance cost, and additional costs (permits, inspections, and other necessary fees). The total cost is calculated using the equation below (20).

$$C_{BSS} = \left(\sum_{n=1}^{N_{BSS}} Invst C_{BSS} \times P_{BSS} \right) + C_{Add} + C_M \times CCF \times N_y \quad (20)$$

- Cost of Power loss (C_{P-loss})

This current study also aims to minimize the power loss cost of the DN. This significantly benefits DSOs by promoting both economic and operational efficiencies. The cost of power loss can be calculated by applying (21) [35].

$$C_{P-loss} = \sum_{y=1}^{N_y} \sum_{h=1}^H P_h^{loss} \times C_{Elec} \times CCF \times D \quad (21)$$

2) TECHNICAL CONSTRAINTS

Equality and inequality constraints are important areas of the power system that are to be carefully sustained to meet efficient system operational performance. To optimally allocate PV and BSS in the DN, the associated constraints are given below.

- Equality Constraints

For secure power system operation, following power balance constraints must be satisfied. [36].

$$P_G(h) = P_D(h) - P_{PV}(h) \pm P_{BSS}(h) + P_h^{loss}(h) \quad (22)$$

- Inequality Constraints

The voltage magnitude of each bus must be maintained between V_{min} and V_{max} set to 0.9 p.u. and 1.1 p.u., respectively [36].

$$V_{min} \leq V_i^h \leq V_{max} \quad (23)$$

The Power and energy of the BSS are also controlled to protect it during charging and discharging processes. The constraints to keep the battery safe during charging and discharging processes are given in (24) to (26). In (24), the state of charge of the BSS constraint is given. To protect BSS lifespan, battery charging and discharging are guided by the maximum SOC (SOC_{max}) and minimum SOC (SOC_{min}). In equation (25) the charging and discharging power processes are protected by setting minimum and maximum values. Similarly, in equation (26) the BSS energy is also protected by the attached set minimum and maximum boundaries [33].

$$SOC_{min} \leq SOC \leq SOC_{max} \quad (24)$$

$$P_{BSS}^{min} \leq P_{chg}^h, P_{dischg}^h \leq P_{BSS}^{max} \quad (25)$$

$$E_{BSS}^{min} \leq E_{BSS}^h \leq E_{BSS}^{max} \quad (26)$$

IV. PROBLEM FORMULATION

The charging behavior of EV charging load in the DN and PV generation are inherently intermittent. To tackle this intermittency, the expected EV charging demand and PV generation is

accurately forecasted using GRU algorithm. The input parameter of the prediction model in this study are energy consumed by the EVs and energy delivered by PV. The forecasted EV demand and PV generation curve are the output parameter of the GRU model. The present study aims to place EVCI in a distributed manner across the DN, rather than concentrating it at a single location, to ensure mutual benefits for both the DSO and EV owners.

The increasing demand on the DN caused by EV presence poses immense challenges to the network's reliability and stability. The optimal allocation of PV and BSS into the DN using a novel EHHO algorithm addressed the EV demand while satisfying the system technical constraints. The input parameters of the optimization problem are electrical network bus data and line data, cost functions, and PV and BSS locations. The optimal number of PV panels, BSS units, optimal capacities of PV, and BSS are the output parameters. The objective functions as given in (17) to (21), in the proposed work, are formulated to maximize the techno-economic benefit of DSO. It is ensured that all the security constraints as given in (22) to (26) are satisfied. The objective function of the present problem is as follows:

1. Minimizing the power loss costs to improve the technical efficiency of the network.
2. Minimizing PV and BSS total costs, comprising both investment, operation, and maintenance costs, addressing the economic aspect

V. RECURRENT NEURAL NETWORK (RNN)-GATED RECURRENT UNIT (GRU)

Conventional RNN is a neural network design with an internal loop to retain previous state memory, making it best suited for time series data processing [37]. However, RNN has an inherent challenge of vanishing gradients, which has led to the development of advanced variants of RNN, such as LSTM and GRU, for the time series forecasting problem. GRU advances RNN architecture by incorporating specialized gating mechanisms that effectively manage hidden state updates and overcome the vanishing gradient problem [38]. The GRU algorithm has a simpler structure with fewer parameters, thereby making it faster than LSTM during training. GRU has two gates, update and reset gate instead of three gates as applied in LSTM [39]. The update gate (z_t) in the GRU model selects important data from past time steps to be retained for future computations. This mechanism effectively allows the GRU model to learn from long dependencies. The output of the update gate is calculated by (27). The reset gate (r_t), on the other hand regulates the quantity of previous information that should be discarded during the training process. The activation function sigmoid (σ) is applied to both the update gate in (27) and the reset gate in (28). Equation (29) computes the candidate hidden state $\hat{h}_t$, where the reset gate r_t controls how much past information is considered. Equation (30) updates the final hidden state h_t by combining past memory and the candidate state, with the update gate z_t balancing retention and

new information.

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z) \quad (27)$$

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r) \quad (28)$$

$$\hat{h}_t = \sigma_h(W_h x_t + U_h (r_t \odot h_{t-1}) + b_h) \quad (29)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \hat{h}_t \quad (30)$$

VI. OPTIMIZATION TECHNIQUE

A detailed description of the traditional HHO [40], MHHO [41], IHHO [35], and the proposed EHHO technique is presented below.

A. TRADITIONAL HHO

In the traditional HHO algorithm, the parameter Escaping Energy plays a vital role in achieving the optimal solution since it determines the transition between the exploration and exploitation phases, as expressed in (31). The program shifts to exploration phase when $E \geq 1$, and shifts to exploitation phase when $E < 1$ [40].

$$E_{conv} = 2E_{in} \times E_1 \text{ And } E_1 = \left(1 - \frac{it}{it_{max}}\right) \quad (31)$$

In the traditional approach, E_{conv} decreases linearly with the number of iterations. However, this linear reduction often disrupts the balance between exploration and exploitation, which restricts global search ability and increases the risk of premature convergence. Thus, modifying the computation of E_{conv} is essential to achieve faster convergence and higher accuracy.

In traditional HHO, the exploration phase can be calculated as follows:

$$X(it+1) = \begin{cases} X_{rd}(it) - r_1 |\Delta X|, & q \geq 0.5 \\ (X_{tar}(it) - X_m(it)) - r_3(lb + r_4(ub - lb)), & q < 0.5 \end{cases} \quad (32)$$

$$\Delta X = X_{rd}(it) - 2r_2 X(it) \quad (33)$$

$$X_m(it) = \frac{1}{G} \sum_{g=1}^G X_g(it) \quad (34)$$

q, r_1, r_2, r_3, r_4 are the random number between 0 and 1. G is the number of hawks.

The exploitation phase can be carried out in four distinct stages depending on the random value (r) and E_{conv} . The values of r ranges between 0 and 1. The detailed description is provided below.

- 1) *Soft besiege* ($r \geq 0.5$ and $|E_{conv}| \geq 0.5$) [40]

$$X(it+1) = [X_{tar}(it) - X(it)] - E_{conv} |JX_{tar}(it) - X(it)| \quad (35)$$

- 1) *Hard besiege* ($r \geq 0.5$ and $|E_{conv}| < 0.5$) [40]

$$X(it+1) = X_{tar}(it) - E_{conv} |X_{tar}(it) - X(it)| \quad (36)$$

- 1) *Soft besiege with progressive rapid dives* ($r < 0.5$ and $|E_{conv}| \geq 0.5$) [40]

$$X(iter + 1) = \begin{cases} WifF(W) < F(X(it)) \\ ZifF(Z) < F(X(it)) \end{cases} \quad (37)$$

- 1) *Hard besiege with progressive rapid dives* ($r < 0.5$ and $|E_{conv}| < 0.5$) [40]

$$X(iter + 1) = \begin{cases} WifF(W) < F(X(it)) \\ ZifF(Z) < F(X(it)) \end{cases} \quad (38)$$

B. MODIFIED HHO (MHHO)

In [41], the escaping energy equation is reformulated to adopt an exponentially decreasing strategy and can be expressed as

$$E^M = 2E_{in} \times E_1^M \text{ and } E_1^M = e^{\left(-\frac{it}{it_{max}}\right)} \quad (39)$$

Nevertheless, [41] does not introduce any modifications to the exploration and exploitation phases of traditional strategy.

C. IMPROVED HHO (IHHO)

The algorithm was improvised by the authors in [35] to develop IHHO. In [35], the escaping energy is further modified as shown in (40). Unlike the traditional linear form as given in (31), the exponential function introduces nonlinearity and adaptability. Moreover, instead of depending on iteration count, convergence is adaptively guided by fitness comparison, enabling faster convergence regardless of the maximum iterations.

$$E^I = 2E_{in} \times E_1^I \text{ and } E_1^I = e^{-(1+\psi)\frac{fit}{fit_{best}}} \quad (40)$$

where ψ is a uniformly distributed random value in the range [1, 0]. Despite modifying escaping energy, [35] retains the traditional exploration and exploitation strategies.

D. ENHANCED HHO (EHHO)

In the present work, the algorithm is advanced from IHHO by improving the exploration phase to enhance solution diversity and prevent premature convergence to local optima. Consequently, the overall performance is strengthened through the combined improvement in escaping energy calculation as in (40) and the exploration phase as in (41)-(46), which is introduced for the first time in the present work. Details of the proposed novel EHHO is given as follows.

Improved Exploration phase:

To enhance the exploration phase and avoid premature convergence, the following modifications are introduced:

For $q \geq 0.5$, the updated position is calculated as

$$X^I(it + 1) = X_{tar}(it) - r'_1 \Delta X', \quad (41)$$

$$\Delta X' = (X_{tar}(it) - 2r'_2 X(it)) \quad (42)$$

r'_1, r'_2 are the random numbers between 0 and 1. $\Delta X'$ indicates the difference in the position between the hawk and the target.

$$C^{lm} = 0.1 \times (ub - lb) \quad (43)$$

$$\Delta X' = \begin{cases} C^{lm}, & \text{if } \Delta X' > C^{lm} \\ -C^{lm}, & \text{if } \Delta X' < -C^{lm} \\ \Delta X', & \text{Otherwise} \end{cases} \quad (44)$$

To accelerate the search towards the optimal solution, the new position is adjusted in alignment with the target position, i.e., the best solution obtained so far. Following this strategy, (32-34) of traditional HHO under the condition $q \geq 0.5$ is refined as expressed in (41-42). Additionally, in the computation of ΔX in (32-33), the modulus operator is removed to allow both positive and negative values, as reflected in (41-42). However, if the value of $\Delta X'$ becomes excessively large or too small, the algorithm may either diverge or exhibit poor convergence performance. Hence, to achieve a balance between diversification and convergence, $\Delta X'$ is constrained in proposed EHHO within an appropriate range, as formulated in (43) and (44).

For condition $q < 0.5$, the updated position is calculated as:

$$X^I(it + 1) = (X_{tar}(it) - r'_3 X_m(it)) - r'_4 (X_m(it) - X(it)) \quad (45)$$

r'_3, r'_4 are the random numbers whose value ranges between 0 and 1.

Using the conventional formulation for condition $q < 0.5$ as given in (32), it is observed that the updated hawk's position tends to shift abruptly relative to the best position, which hinders the convergence speed. To overcome this limitation and accelerate the search process, an improved formulation is introduced in (45). The proposed formulation is designed to mitigate abrupt variations between the updated hawk positions and the current best solution. This adjustment promotes smoother position updates and enables a more efficient progression toward the global optimum. Accordingly, the modified exploration-phase update can be expressed as follows:

$$\begin{aligned} X^I(it + 1) &= \begin{cases} X_{tar}(it) - r'_1 (X_{tar}(it) - 2r'_2 X(it)), & q \geq 0.5 \\ (X_{tar}(it) - r'_3 X_m(it)) - r'_4 (X_m(it) - X(it)), & q < 0.5 \end{cases} \end{aligned} \quad (46)$$

No improvement is applied to the exploitation phase and can be calculated using (35)-(38).

E. BENCHMARK TEST FUNCTION

The performance of the proposed EHHO algorithm is validated using three well-known benchmark functions, namely the Sphere, Rastrigin, and Ackley functions. The obtained results are compared against those of other metaheuristic approaches, including IHHO, MHHO, traditional HHO, PSO, and GA to demonstrate the superiority and robustness of the proposed method.

For all algorithms, the population size, number of variables, and maximum iterations are set to 200, 50, and 200, respectively. Each benchmark function is executed over

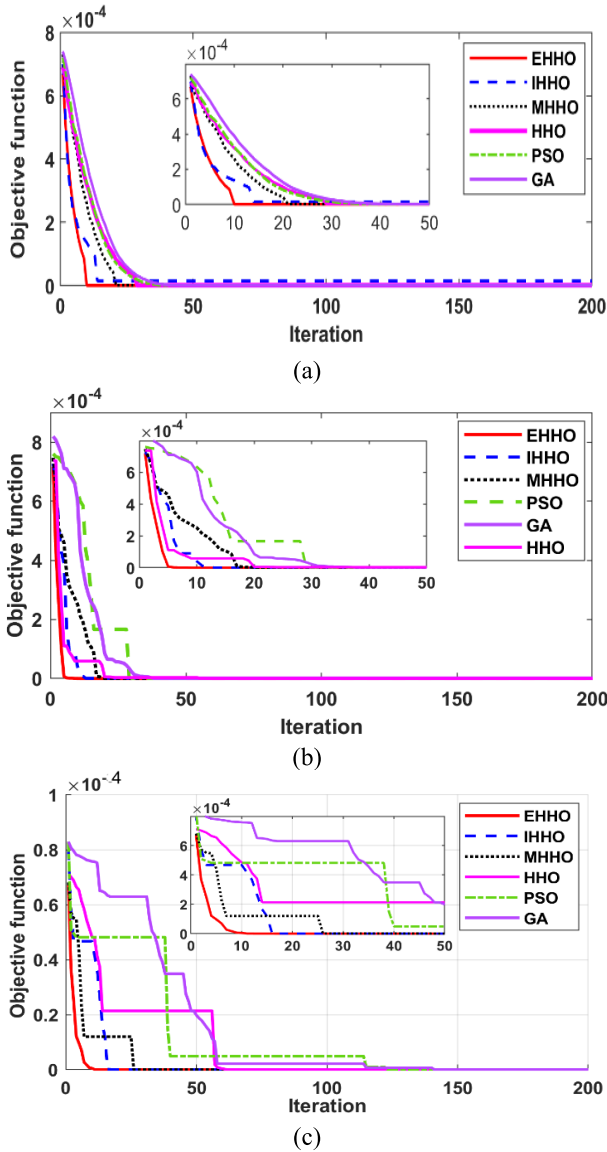


FIGURE 1. Convergence comparison of different algorithms for benchmark functions: (a) Sphere function, (b) Rastrigin function, (c) Ackley function.

30 independent runs to prove robustness. Fig. 1 indicates the comparative convergence characteristics between EHHO, IHHO, MHHO, HHO, PSO, and GA for all the three benchmark test function. As illustrated in Fig. 1, EHHO converges to the global optimum with significantly fewer iterations, thereby validating its efficiency and robustness.

VII. METHODOLOGY

This section elaborates on the methods employed in this study to evaluate the evolving problems associated with EV consumption in the DN.

A. IMPLEMENTATION OF GRU FOR EV CHARGING DEMAND AND PV GENERATION FORECASTING

Fig. 2 presents a detailed breakdown of the steps involved in the EV charging demand forecasting with the GRU

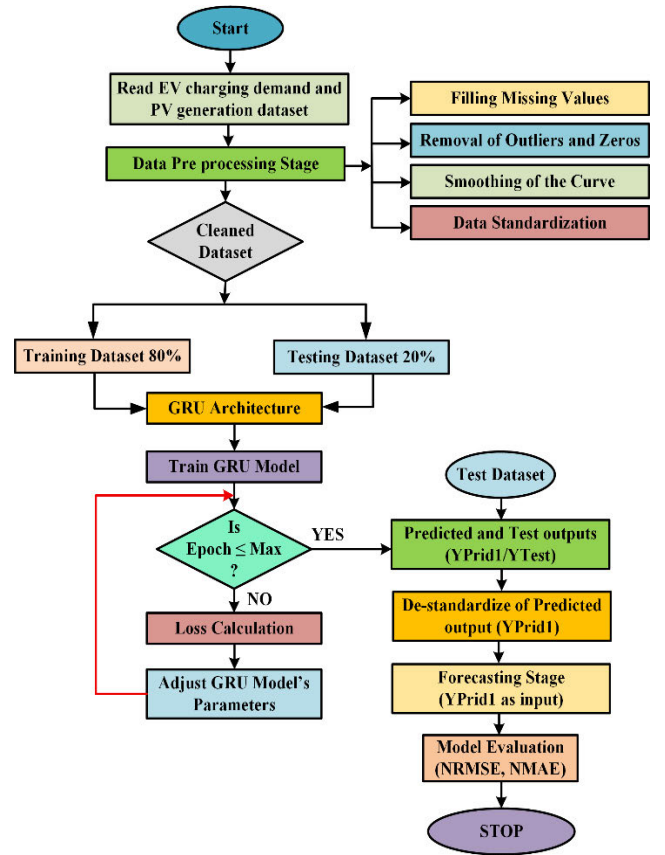


FIGURE 2. EV load and PV generation forecasting procedure.

model. Beginning with data preprocessing, followed by the requisite parameter setting. The model is first subjected to training and subsequently evaluated through the testing dataset.

1) DATA PREPROCESSING FRAMEWORK

The historical charging dataset used for forecasting originates from Caltech's repository [42] and is stored in a comma-separated values (CSV) format. Initially, the raw data contained inconsistencies like missing values, outliers, and zeros, which could compromise the accuracy of the model. The 'Data Clean APP' in MATLAB R2024b is used for the preprocessing. The standardization of the data is achieved via the Z-score method [47] [39].

$$X_{std} = \frac{x - \text{mean}}{\text{standard deviation}} \quad (47)$$

The proper processing of the data enhances the accuracy of the subsequent estimation processes. After the standardization of the data, the data is split into training (80%) and testing (20%) dataset [39].

2) MODEL EVALUATION

Evaluating the performance of the GRU model, the RMSE and the MAE matrices are adopted in this study [43].

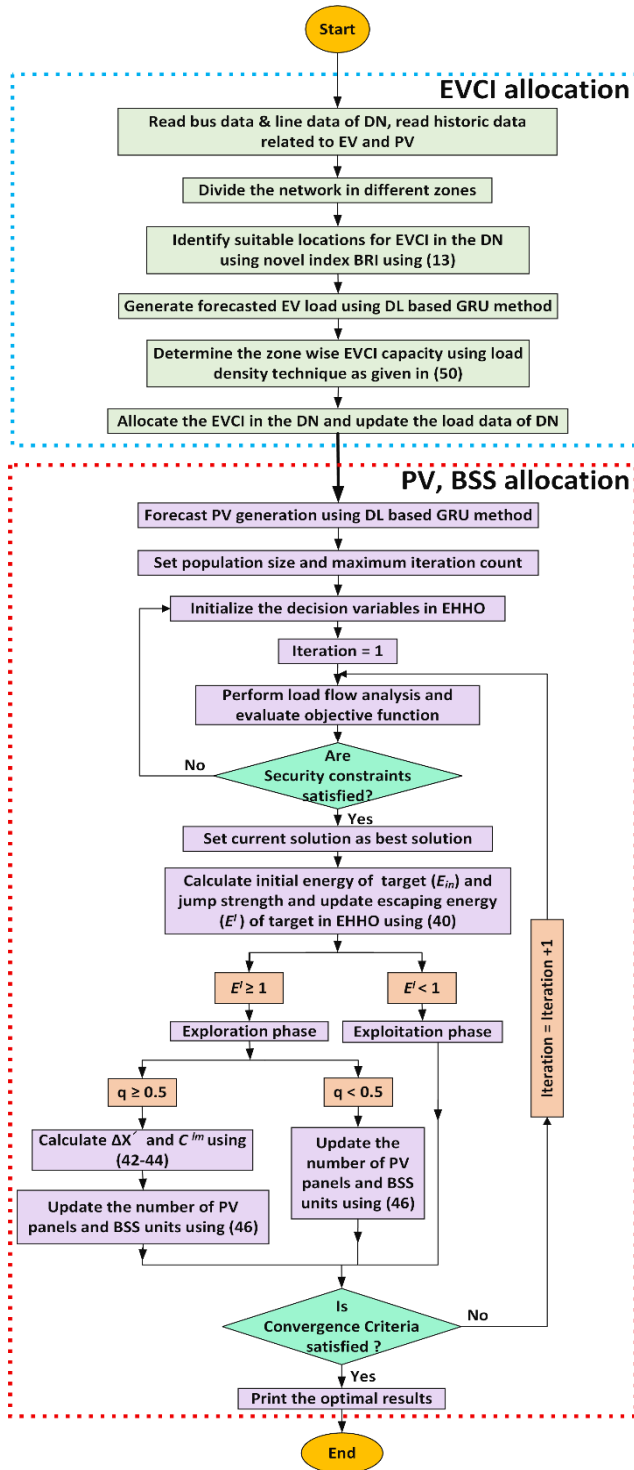


FIGURE 3. Flowchart of the implementation of novel EHHO algorithm.

The RMSE measures the average difference between the predicted values (Y_i) and actual values ($\hat{Y}_i$), providing understanding of the model's accuracy. The Normalized Mean Absolute Error (NMAE) matrix, on the other hand, calculates the average magnitude of the prediction errors. NRMSE and

TABLE 3. GRU Structure Parameter Setting.

Parameter Type	Parameter Values
Input Size	1
Number of Responses	1
Number of Hidden Layers	200
Maximum Epochs	100
Initial Learning Rate	0.005
Dropout rate	0.2
Optimizer	Adam
Activation function	Sigmoid

NMAE [39] can be calculated as follows.

$$NRMSE = \frac{\sqrt{\frac{1}{N} \sum_{i=1}^n (\hat{y} - x)^2}}{x_{max} - x_{min}} \quad (48)$$

$$NMAE = \frac{\frac{1}{N} \sum_{i=1}^n |\hat{y} - x|}{x_{max} - x_{min}} \quad (49)$$

B. EVCI LOCATION AND SIZING

In the present work, the DN is divided into zones to ensure a balanced allocation of EVCI across the planning area. This zonal division strategy reduces EV owners' stress of traveling long distances to recharge their EVs. It also aids DSO in distributing the CI within the network, preventing the concentration of EVCI at one place. The suitable location for EVCI placement in each zone of DN is then identified using novel index BRI. The BRI evaluation identifies the strong buses with low BRI values, and weak buses with high BRI values in both networks. The strong buses in each zone of the network are then identified for CI placement based on the low BRI values of that zone.

Further, EV charging demand is forecasted using DL based GRU model. Subsequently, the load density method [44] is employed to determine the CI capacity at the identified bus location in each zone. The total load of the network accounts for both residential demand and forecasted EV charging demand. The CI loads are distributed across the allocated zones into which the DN is divided. Applying the load density method (50), the CI capacity, S_{CI} ($CI = 1, 2, \dots, n$), is determined at each identified bus in the DN where CI is placed. The charging simultaneity (K_s) of the EV is given as 0.2 [44] in this current study. CI capacity per zone can be determined using (50).

$$S_{CI} = K_s \times N_{EV} \times P_{ch} \times \frac{L_{zone}}{L_{total}} \quad (50)$$

C. IMPLEMENTATION OF EHHO ALGORITHM FOR OPTIMAL ALLOCATION OF PV AND BSS

In this study, novel EHHO technique is used to optimally allocate PV and BSS in the DN. The objective function (20) of the optimization algorithm minimizes the system's overall costs, constituting the PV cost (22), BSS cost (23), and power

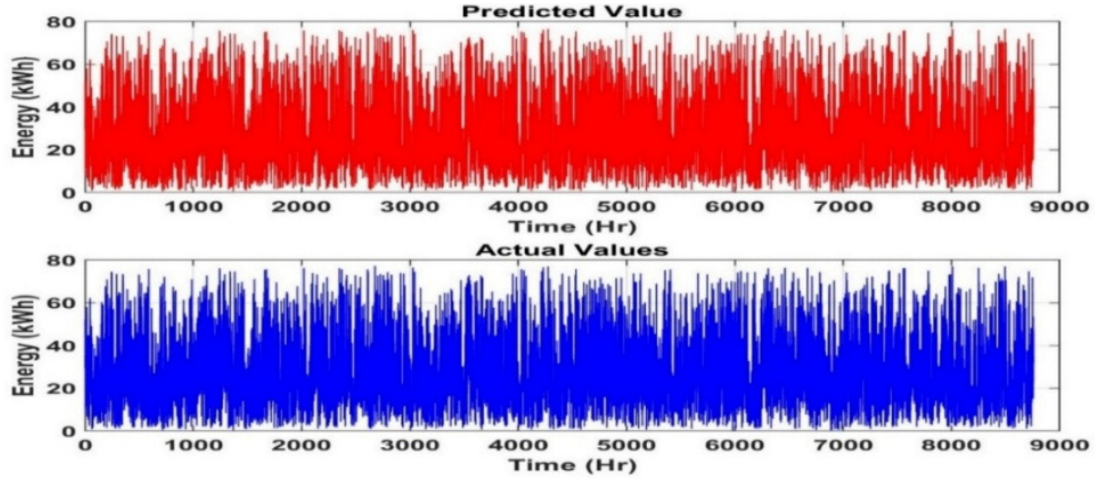


FIGURE 4. Comparison of Predicted and Actual Values of EV load obtained through GRU.

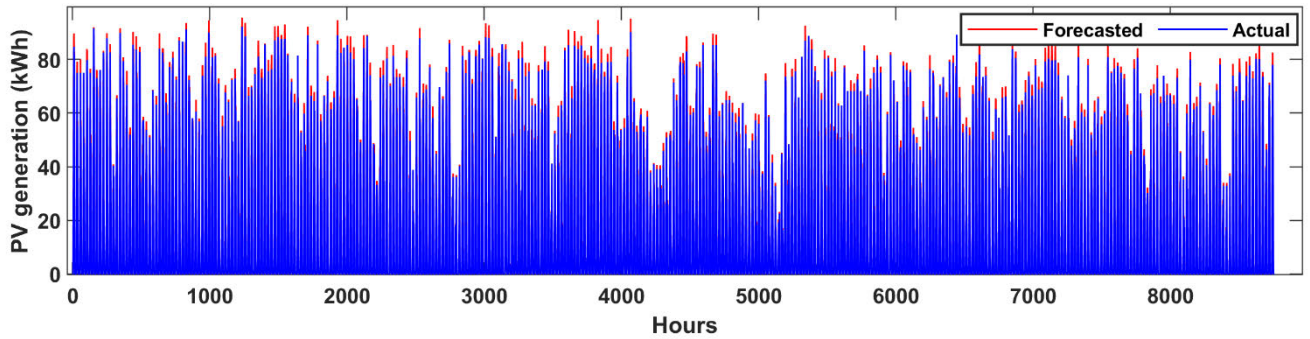


FIGURE 5. Comparison of Predicted and Actual Values of PV generation obtained through GRU.

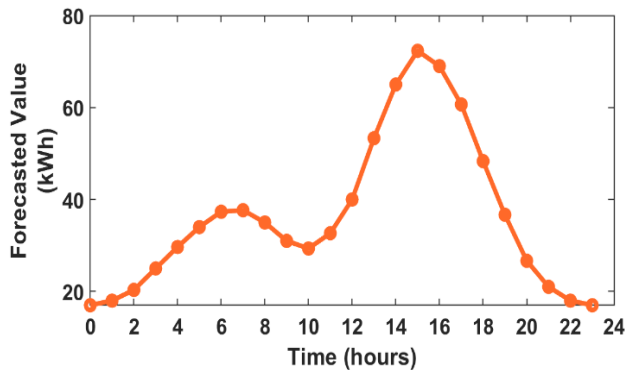


FIGURE 6. EV load forecasted values over a 24-hour time interval.

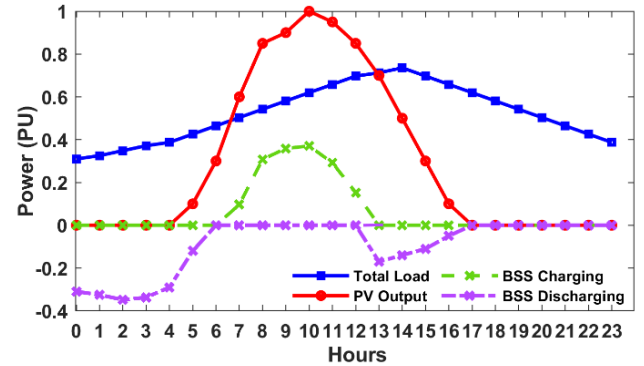


FIGURE 7. Hourly variation of solar power output, BSS charging-discharging power and grid demand in a typical sunny day.

loss cost (24), while considering the system's technical constraints (25) to (29). The technical constraints of the system in this study cover the power flow constraint, the voltage magnitude of the system, and the devices (PV and BSS) associated constraints. The decision variable of the optimization is the number of PV panels and BSS units. Fig. 3 presents the flowchart of the EHHO algorithm used to determine the optimal sizes of PV and BSS. The Figure displays the step-by-step procedures for solving the optimization problem using the EHHO algorithm.

VIII. RESULTS AND DISCUSSIONS

This paper develops a comprehensive planning framework for DN that integrates EVCI, PV, and BSS. The framework aims to minimize the techno-economic and reliability impacts of these integrations on the grid while enhancing the system performance.

A. DATA DESCRIPTION

The present planning framework is tested on the IEEE 15-bus DN and the 28-bus DN [45]. The 15-bus DN comprises

TABLE 4. PV Modules Parameters [31], [34], [35].

PV Module Economic Parameters	
Parameter	Values
Lifetime of PV	10 years
IF, IR	5 %, 10 %
C_{Elec}	5.06 INR/kWh
HOC of PV	0
$InvstC_{PV}$	89972 INR/kWh
C_M	1% of $InvstC_{PV}$
P_{PV}	10 kW

TABLE 5. BSS Unit Parameters [31], [34], [35].

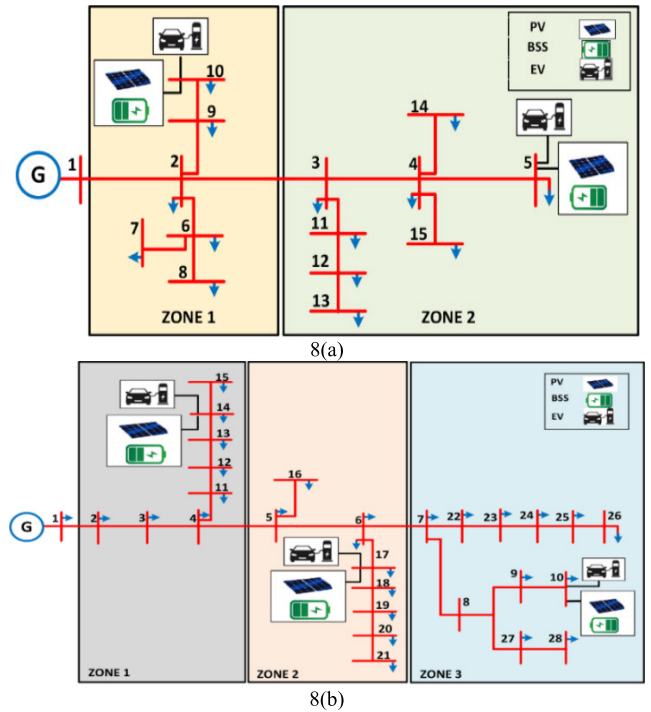
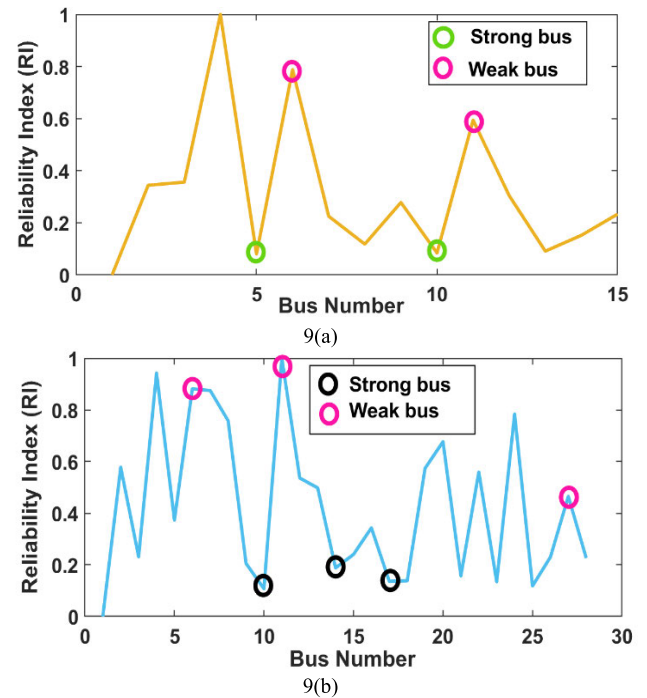
BSS Economic Parameters	
Parameter	Values
$InvstC_{BSS}$	18,177 INR/kW
P_{BSS}	4.8 kW
C_M	1% $InvstC_{BSS}$
C_{Add}	17,766.98 INR
η_{chg}	90%
η_{dischg}	80%

TABLE 6. EHHO Algorithm Parameters.

Parameters	Value
nPop	100
it_{max}	100
lb of PV panels	5
ub of PV panels	20
lb of BSS units	3
ub of BSS units	15

of 14 load buses and 14 lines, with 11 kV and 100 MVA base values. The total active and reactive power demands are 1226.4000 kW and 1251.1785 kVar, respectively. The 28-bus system consists of 27 load buses and 27 lines, with an 11kV and 100 MVA base value. The total active and reactive power demands are 761.0400 kW and 776.4190 kVar, respectively [45]. The historical EV charging load dataset used in this study is collected from the data repository of the California Institute of Technology (Caltech), based in Pasadena, California, USA [42]. The dataset is formatted as a CSV file titled 'SYNTHETIC_EV_DATA.csv'. The Excel file contains four important columns: 'Connection Time', 'Charging Duration', 'kWh-Delivered', and 'Day Indicator'.

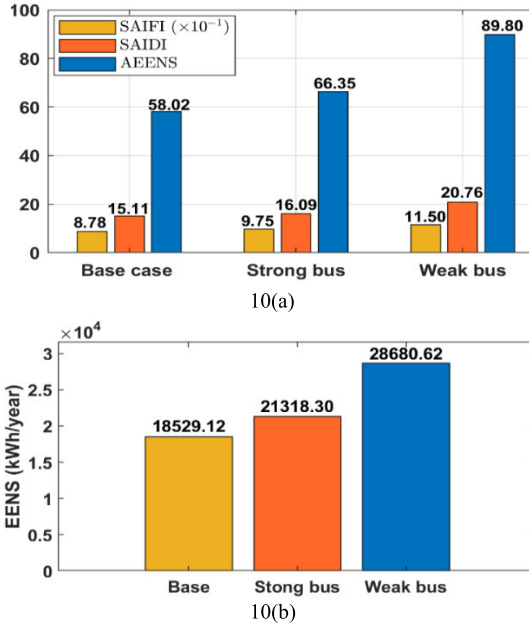
The 'Connection Time' column records the time each EV got connected to the grid in a 24-hour interval. The second and third columns, 'Charging Duration' and 'kWh-Delivered,' detail the charging duration in hours and quantify the energy delivered in kWh, respectively. Finally, the 'Day Indicator' column denotes the specific day the data was recorded. The dataset in the CSV file comprises 1,965,239 rows, representing the EV charging record for 29,600 days. In this study, from the overall quantity of the

**FIGURE 8.** Network diagrams of (a) IEEE 15-bus, (b) Indian 28-bus system.**FIGURE 9.** BRI values for (a) IEEE 15-bus, (b) IEEE 28-bus system.

dataset collected, a 5-year sample EV charging dataset is used for the EV load forecasting [46]. A total of 37 EVs are considered in the present work [46]. The PV generation forecasting in this study is carried out using data obtained from a solar power plant located in Karnataka, India,

TABLE 7. CIs Capacity Sizing and Allocation.

IEEE 15-BUS DN		
Bus No.	Zone	CI Capacity (kW)
5	2	43.33561644
10	1	30.66438356
28-BUS INDIAN RURAL DN		
Bus No.	Zone	CI Capacity (kW)
10	3	25.3201
14	1	18.1869
17	2	30.4930


FIGURE 10. Results of (a) SAIFI, SAIDI, AEENS, and (b) EENS for IEEE 15-bus system.

covering a period of four years [47]. The dataset includes date, time, and corresponding hourly PV energy generation. For model development, PV energy generation has been employed as the primary input feature to the GRU. Table 3 shows details of the entire configuration parameters for the GRU model construction.

The comparison between the predicted and actual values for both EV load and PV generation is illustrated in Fig. 4 and Fig. 5, respectively. The graph illustrates high predictive accuracy for the GRU model, with predicted values closely aligning with the actual values, with negligible differences observed in the output. The forecasted result of the EV charging load over 24 hours is shown in Fig. 6, revealing two peak consumption times. The first occurs between 4:00 hour and 9:00 hour, and the second peak occurrence is observed between 13:00 hour and 17:00 hour. The first peak period links to EV owners preparing for their daily travel jobs, while the second peak occurs during work hours and other engagements, at which time the CIs experience a significant increase in charging activity. However, the maximum peak occurs at 15:00 hour, with consumption of 74 kW.

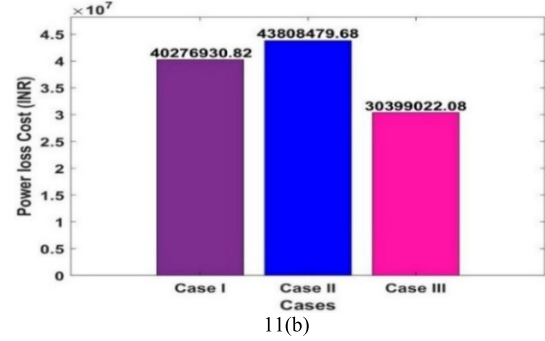
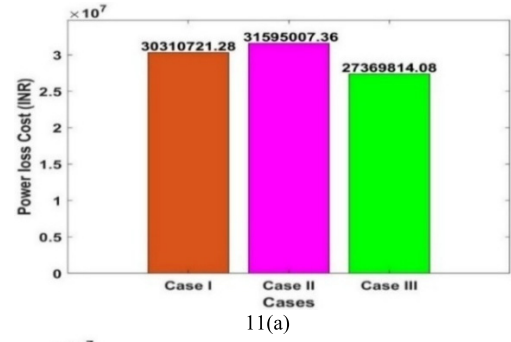
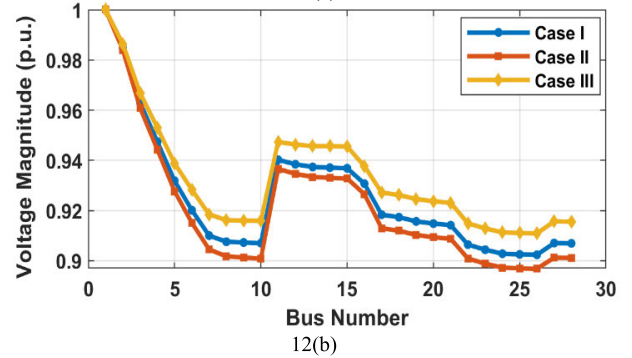
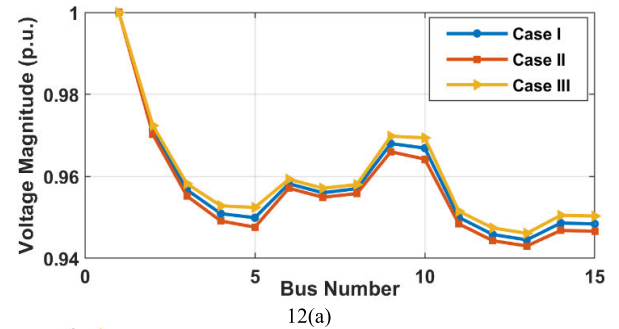

FIGURE 11. Comparison of power loss cost across all cases for (a) 15-bus system and (b) 28-bus system.

FIGURE 12. Comparison of voltage profile across all cases for (a) 15-bus system and (b) 28-bus system.

Fig. 7 indicates the forecasted hourly PV generation, in a typical sunny day and hourly demand variations of the system [48]. The hourly operation of BSS is also illustrated in Fig. 7. As observed from Fig. 7, the peak PV generation is occurred at 11:00 hour while the peak total demand is

TABLE 8. Optimal Size and Location of PV and BSS in the DNs for 15-bus and 28-bus DN.

IEEE 15-BUS DN					
Bus No.	Zone	Number of PV panels	PV Capacity (kW)	Number of BSS units	BSS Capacity (kWh)
5	2	8	80	5	19.20
10	1	9	90	6	28.80
28-BUS INDIAN RURAL DN					
Bus No.	Zone	Number of PV panels	PV Capacity (kW)	Number of BSS units	BSS Capacity (kWh)
10	3	9	90	9	43.20
14	1	7	70	10	48.00
17	2	8	80	9	43.20

TABLE 9. Optimal Cost Values for both 15-Bus And 28-DN.

Sl. No	System	Power loss // cost (INR)	PV Cost (INR)	BSS Investment cost (INR)	Total Cost (INR)
1	15- Bus	27.4×10^6	15.3×10^6	10.1×10^5	43.7×10^6
2	28-Bus	30.4×10^6	21.6×10^6	34.7×10^5	55.5×10^6

occurred at 15:00 hour. It is further observed that after meeting the load, the excess PV power charges the BSS between 6:00 and 13:00 hours. During 0:00 to 5:00 hours, when PV power is unavailable, the BSS discharges to meet the demand. Tables 4-5 presents parameters of the PV and BSS units. These parameters are used in modelling the PV and BSS while optimally allocating the PV and BSS in the DN. Table 6 provides the parameter related to the EHHO algorithm.

B. CASE DESCRIPTION

To adequately assess the network's performance, this paper properly evaluates the techno-economic performance and reliability of both 15-bus and 28-bus DN. The study investigation is conducted in multiple cases: Case I, Case II, and Case III.

- Case I- Base case
- Case II- EVCI placement
- Case III- EVCI placement with PV and BSS

In Case I, the analysis is performed without integrating the EVCI, PV, and BSS into the network. In Case II, the network is divided into zones, and level-2 AC chargers with power rating of 10 kW is integrated into the network [49]. In Case III, the EVCI, PV, and BSS systems are all incorporated into the grid.

C. NUMERICAL RESULTS

This section presents the technical, reliability, and economic results obtained from the proposed planning strategy across all considered cases.

1) RESULTS OF EV LOCATION AND SIZING IN THE DN

The 15-bus network is divided into two zones, while the 28-bus system is divided into three zones, as illustrated in Fig. 8(a) and 8(b). The suitable locations for the CIs allocation are determined by analyzing individual bus reliability using the novel index BRI as given in (11). The BRI results for the 15-bus and 28-bus systems are presented in Fig. 9(a)

and 9(b), respectively. Based on the BRI values, bus no. 5 in Zone 1, and bus no 10 in Zone 2 of 15-bus system identified as strong buses. For 28-bus system, bus no 14 in Zone 1, 17 in Zone 2 and 10 in Zone 3 identified as strong bus. On the other hand, bus no 6, 11 in 15-bus DN and bus nos. 6, 11 and 27 are considered as weak buses across all the zones. Therefore, for 15-bus DN, bus nos. 5, 10 and for 28-bus DN, bus nos. 10, 14 and 17 are selected for EVCI placement. Table 7 presents the CI capacity results, obtained using the load density method as given in (46), for both the IEEE 15-bus and 28-bus DN.,

2) RELIABILITY EVALUATION IN STRONG AND WEAK BUS FOR EVCI ALLOCATION

The effectiveness of placing EVCI at strong buses in the DN is evaluated by comparing the results with those obtained from placements at weak buses, using different reliability indices as performance measures. From Fig. 10(a) and 10(b), it is evident that allocating the EVCI at a weak bus result in higher values of reliability indices such as SAIFI, SAIDI, AEENS, and EENS compared to allocation at a strong bus.

As observed in Fig. 10(b), the EENS is reduced by 26% when the EVCI is allocated at a strong bus instead of a weak bus. Placing CI at either strong or weak buses leads to some deterioration in reliability; however, the rate of deterioration is higher when CI is allocated to weak buses of the DN. This demonstrates the effectiveness of the proposed BRI in guiding CI allocation in the DN to enhance overall system performance.

3) RESULTS OF PV AND BSS SIZE AND LOCATION

Ensuring a continuous power supply regardless of the intermittent nature of the PV power during operation, this study integrates BSS to support the grid by continuously supplying power to the customers during the inactive period of the PV. The PV and BSS sizing allocations in the DN are optimally determined using the EHHO. PV and BSS are allocated at the

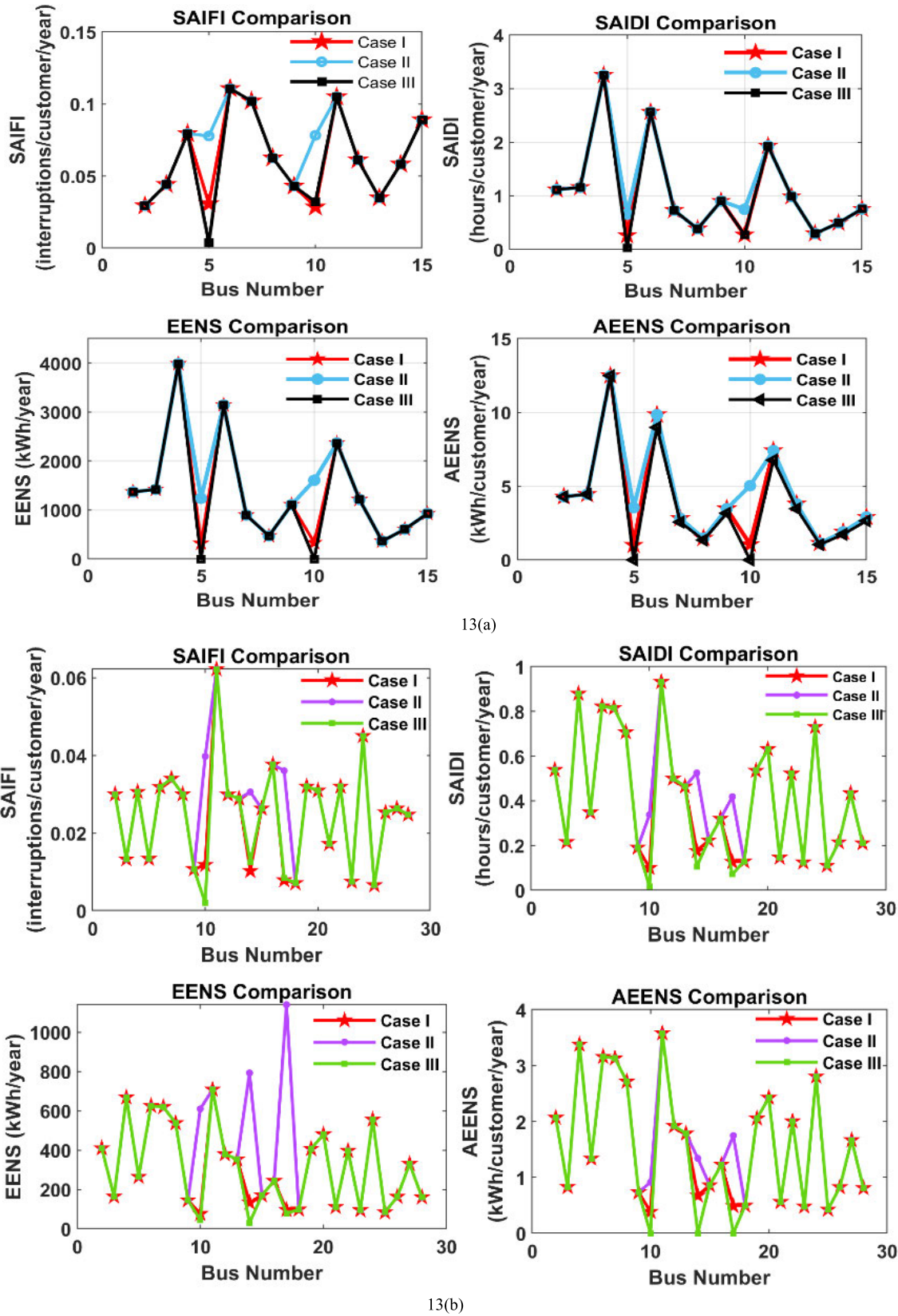


FIGURE 13. Bus wise reliability indices across all cases for (a) 15-bus system and (b) 28-bus system.

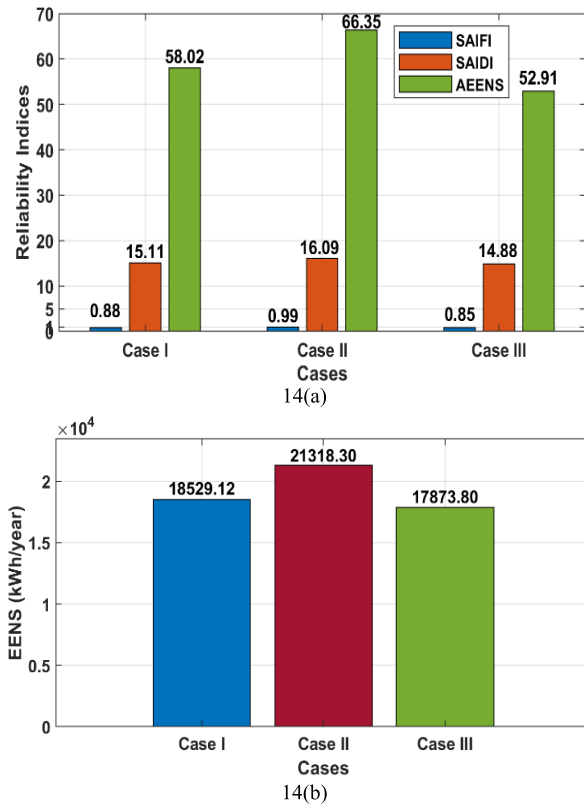


FIGURE 14. Results of (a) SAI, SAID, AEENS and (b) EENS across all cases for 15-bus system.

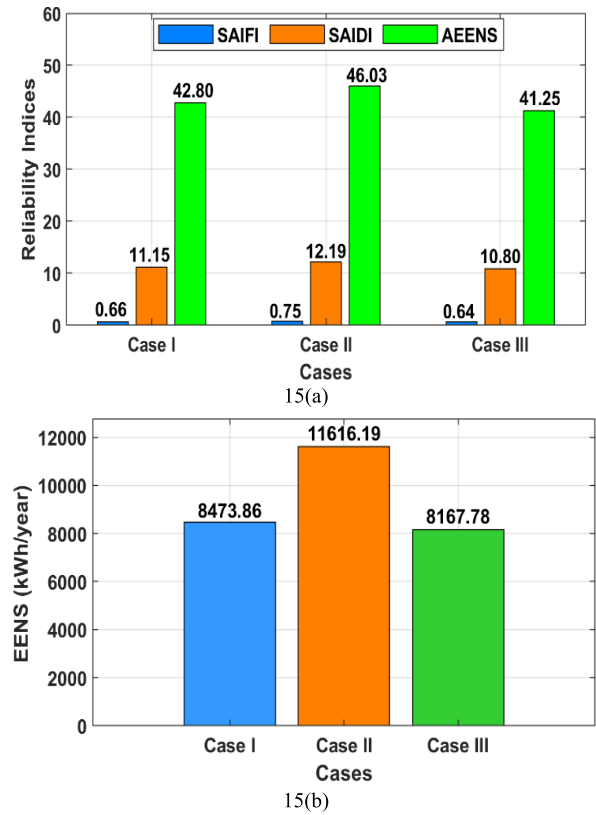


FIGURE 15. Results of (a) SAI, SAID, AEENS and (b) EENS across all cases for 28-bus system.

same location as the CI in the DNs, as displayed in Fig. 8(a) and 8(b). The optimal results of both PV and BSS system capacities are presented in Table 8. In addition, the minimized objective function considering the technical constraints for both test systems are presented in Table 9.

D. COMPARATIVE PERFORMANCE ANALYSIS

In this section, a comprehensive analysis is made considering the techno-economic and reliability performances of the networks across all cases. In Case I, under the base loading condition, the power loss for the 15-bus DN and 28-bus DN is 61.788 kW and 82.104 kW, respectively. When only EVCI is integrated into the DN in Case II, the corresponding losses are 66.062 kW and 91.701 kW for the 15-bus and 28-bus systems, respectively.

The minimum power loss is obtained in Case III, where PV and BSS are integrated along with EVCI, yielding 55.793 kW for the 15-bus and 77.803 kW for the 28-bus system. Consequently, the minimum power loss cost is obtained in Case III as shown in Fig. 11(a) and 11(b). In Case III, the power loss cost decreased by 14% and 30% for 15-bus and 28-bus DN respectively as compared to Case II.

The voltage profiles of the 15-bus and 28-bus DNs for all three cases are shown in Fig. 12(a) and Fig. 12(b), respectively. The integration of only EVCI results in a deterioration of the voltage profile compared to the Case I. In contrast, the

inclusion of PV and BSS in the EVCI-integrated network in Case III significantly improves the voltage profile relative to Case I and Case II.

The system reliability indices, such as SAI, SAID, EENS, and AEENS, are assessed across all the cases to evaluate the system performance. The bus-wise reliability evaluation results for 15-bus DN and 28-bus DN are illustrated in Fig. 13(a) and 13(b), respectively. According to Fig. 13(a) and Fig. 13(b), the reliability (SAI, SAID, EENS, and AEENS) deteriorates at buses 5 and 10 of the 15-bus system and at buses 5, 10, and 17 of the 28-bus DN compared to Case I due to the integration of EVCI. Furthermore, the inclusion of PV and BSS in Case III significantly enhances the reliability of these buses.

The results of the entire system reliability assessment in terms of SAI, SAID, and AEENS AND EENS are illustrated in Fig. 14(a) and 14(b) and Fig. 15(a) and 15(b) for both the 15-bus and 28-bus systems, respectively. The maximum deterioration in reliability is observed in Case II due to the integration of EVCI into the DN without renewable and storage support. In contrast, the integration of PV and BSS in Case III significantly improves the reliability indices compared to Cases I and II. As shown in Fig. 14(b) and Fig. 15(b), EENS improves by 16% and 27% in Case III compared to Case II for the 15-bus and 28-bus systems, respectively. This demonstrates that the presence of PV and

BSS stabilizes the network and enhances system reliability despite the incorporation of CIs.

E. COMPARISON OF DIFFERENT ALGORITHMS

In this section, a comparative analysis of GRU and novel EHHO is carried out against other well-known techniques for solving the proposed problem.

1) VALIDATION OF GRU FOR EV CHARGING DEMAND AND PV GENERATION FORECASTING

The GRU model is compared with RNN, LSTM, and DNN [50] to validate its accuracy in forecasting EV demand and PV generation. The parameters used for RNN, LSTM and DNN are same as GRU. For 24-hour load forecasting, the errors are evaluated against these models using RMSE and MAE as error metrics, summarized in Table 10. During both training and testing, the GRU model outperforms the other techniques, achieving the lowest NRMSE of 1.52% and 2.92%, and the lowest NMAE of 0.48% and 0.78%, respectively. These results confirm the superior performance of GRU over other machine learning techniques for EV load and PV generation forecasting.

2) VALIDATION OF PROPOSED NOVEL EHHO TECHNIQUE

To validate the efficacy of the proposed EHHO technique for the present optimization problem, its performance is compared against existing soft computing methods, including IHHO, MHHO, traditional HHO, PSO, and GA. Fig. 16(a) and Fig. 16(b) illustrate the convergence curves of EHHO and other algorithms for both the 15 bus and 28-bus networks, respectively. The results demonstrate that the EHHO algorithm achieves the global optimum with fewer iterations, thereby exhibiting superior convergence efficiency. Table 11 and Table 12 present the comparative analysis of algorithm consistency over 30 trials for the 15-bus and 28-bus DN, respectively. The results show that the proposed EHHO technique exhibits the smallest gap between maximum, minimum, and average values, indicating high consistency for both the test system, establishing the efficacy of EHHO.

In contrast, GA demonstrates the largest variation between minimum, maximum and average values, highlighting the superior consistency of EHHO. This enhanced performance is primarily attributed to its improved escaping energy mechanism and exploration capability, which enable it to outperform other algorithms forecasting using real-world data.

3) SENSITIVITY ANALYSIS

To examine the robustness of utilizing the BRI method for strong and weak bus estimation under different conditions, a sensitivity analysis is carried out by varying the load density by $\pm 20\%$ from its base value for both 15-bus and 28-bus network. The results of sensitivity analysis for varying load density are given in TABLE 13. Fig. 17 and Fig. 18 represent the strong and weak bus obtained through BRI values for 15-bus and 28-bus system for varying load density, respectively.

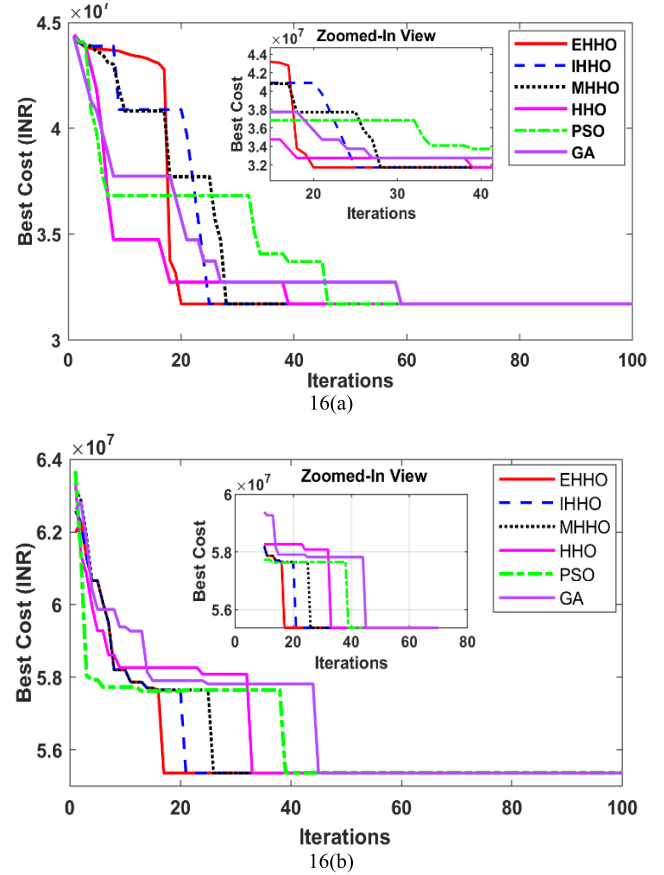


FIGURE 16. Convergence characteristics of different algorithms for (a) IEEE 15-bus DN and (b) 28-bus DN.

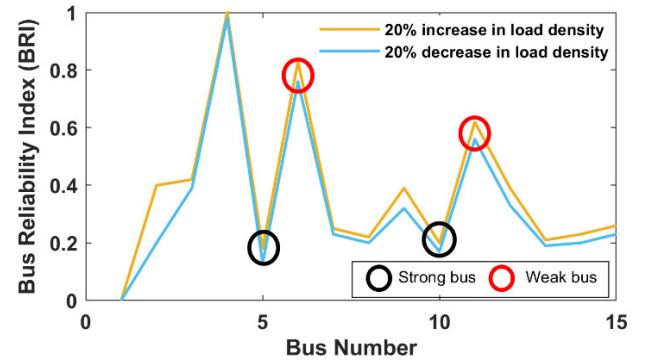


FIGURE 17. Sensitivity analysis of BRI by changing load density for 15-bus system.

From TABLE 13 it is inferred that the BRI value varies proportionally with the load density. An increase in load results in a higher BRI value, indicating a reduction in bus reliability, whereas a decrease in load leads to a lower BRI value, signifying improved reliability. A lower BRI value corresponds to a strong bus, while a higher BRI value represents a weak bus. However, despite the variations in the absolute BRI values with changes in load density, the same buses continue to be identified as strong and weak as those

TABLE 10. Comparison Of GRU, RNN, LSTM, and DNN Performance.

Model	Training NRMSE (%)	Testing NRMSE (%)	Training NMAE (%)	Testing NMAE (%)
GRU	1.52	2.91	0.48	0.78
LSTM	1.61	2.96	0.84	1.19
RNN	1.71	3.36	0.79	1.11
DNN	1.97	3.79	0.50	0.94

TABLE 11. Comparison Of consistency of different soft computing techniques after 30 trials for 15-bus DN.

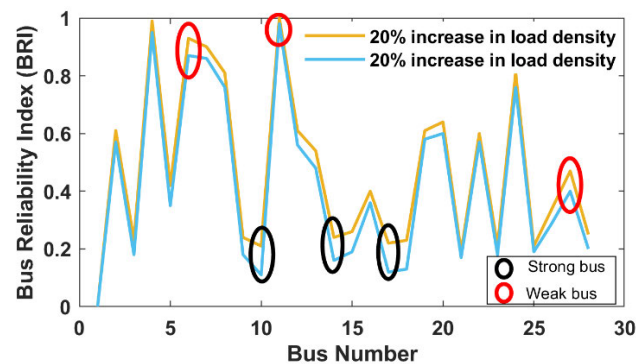
Performance indices	EHHO	IHHO	MHHO	HHO	PSO	GA
Maximum	4.379×10^7	4.381×10^7	4.383×10^7	4.389×10^7	4.392×10^7	4.393×10^7
Minimum	4.363×10^7	4.364×10^7	4.364×10^7	4.362×10^7	4.357×10^7	4.358×10^7
Average	4.370×10^7	4.371×10^7	4.373×10^7	4.374×10^7	4.372×10^7	4.371×10^7

TABLE 12. Comparison Of consistency of different soft computing techniques after 30 trials for 28-bus DN.

Performance indices	EHHO	IHHO	MHHO	HHO	PSO	GA
Maximum	5.556×10^7	5.560×10^7	5.562×10^7	5.565×10^7	5.572×10^7	5.574×10^7
Minimum	5.548×10^7	5.549×10^7	5.549×10^7	5.551×10^7	5.556×10^7	5.557×10^7
Average	5.551×10^7	5.553×10^7	5.554×10^7	5.556×10^7	5.562×10^7	5.562×10^7

TABLE 13. Sensitivity Analysis on Changing System Load Density.

Change in load density	Minimum BRI value	
	15-bus system	28-bus system
- 20%	0.13	0.11
0%	0.14	0.16
+ 20%	0.18	0.21

**FIGURE 18.** Sensitivity analysis of BRI by changing load density for 15-bus system.

obtained under the base load density condition as observed in Fig. 16 and Fig. 17. Overall, the findings confirm the model's sensitivity and responsiveness under varying conditions.

IX. CONCLUSION

This study offers a comprehensive planning framework considering EVCI, PV systems, and BSS integration in the DN, focusing on the enhancement of the technical, reliability, and economic performance of the grid. Addressing the inherent uncertainty in EV charging demand and PV generation on the DN, a DL-based GRU model was developed and the its efficacy was established using error metrics such as NRMSE

and NMAE. Utilizing a novel reliability-based index BRI, the zone wise allocation of EVCI was done. Further, the efficacy of the BRI for EVCI allocation was validated as described in Section VIII.

For optimal allocation of PV and BSS in EVCI integrated DN, a novel EHHO technique was applied considering minimization of power loss cost, installation, operation and maintenance cost of PV and BSS. All the security constraints were satisfied during the optimization process. It was established that the application of the novel EHHO achieved superior results when tested on both benchmark functions and the present problem, demonstrating higher accuracy and faster convergence compared to other techniques.

The proposed methodology was tested on the IEEE 15-bus and the 28-bus Indian rural DNs over a period of 10 years. To get a thorough insight into the proposed methodology, different case studies were adopted in the present work. The results revealed that the integration of PV and BSS in the DN leads to notable improvements in reliability indices (SAIFI, SAIDI, EENS, and AEENS) for both test systems. Moreover, the inclusion of PV and BSS significantly enhances both bus-wise reliability and the entire system reliability. In particular, the presence of PV and BSS alongside EVCI reduces the EENS by 16% in the 15-bus DN and by 27% in the 28-bus DN, compared to the case with only EVCI integration. It was further observed that, EVCI integration alone in the DN degrades voltage and increases losses, whereas PV and BSS integration improves both voltage profile and power loss.

Overall, the proposed framework demonstrates significant improvements in the reliability, technical and economic performance of DN with EVCI, PV, and BSS integration. Future work will focus on the implementation of V2G technology along with reliability analysis to further enhance the performance of the DN.

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